# OpenCaseLaw: A Verifiable Multilingual Citation Graph for Swiss Jurisprudence


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**Abstract**

We release **OpenCaseLaw**, an open, verifiable retrieval substrate for Swiss case law: a resolved cross-court citation graph bridged to statute articles, Federal Council legislative-history materials, and scholarly commentaries, with a Swiss-native identifier scheme and cryptographic provenance. The release contains **972,882** court decisions across **109 courts** in 28 jurisdictional layers (DE/FR/IT, 1875–2026), **8.05 M** resolved citation tokens (92.9 % coverage of 8.66 M extracted tokens) expanded to **8.10 M** (source-decision, target-decision) link edges, **11.28 M** decision-to-statute edges across **5,516** federal and **15,722** cantonal laws, **362** CC-BY/CC-BY-SA commentaries, and **8,124** article-level Materialien anchors. Every decision carries a Swiss-native `cli:ch` identifier with an ECLI projection and a cryptographic membership proof against a 2026-05-21 RFC-6962/OpenTimestamps integrity root. The corpus runs as a live MCP/REST endpoint at `mcp.opencaselaw.ch`. Quality controls: recorded-date chronology violations on 1.53 % of edges, implying a date-sanity-visible precision ceiling of 98.47 % rather than a lower bound; a 400-sample rule-based consistency check (400/400 pass; Wilson 95 % CI [99.0 %, 100.0 %] for the checked property). Corpus CC0 on Hugging Face; code MIT. Snapshot 2026-05-21.


## 1 Introduction

Swiss court decisions are published, but the publishing landscape is fragmented. Each of 109 courts across 26 cantons and the federal layer maintains its own portal, with its own format, fields, and refresh cadence; no public surface aggregates them as a queryable whole. Swiss legal NLP has therefore lacked a public substrate joining the objects legal practitioners actually move between: court decisions, case citations, statute articles, legislative materials, doctrinal commentary, identifiers, and provenance. Existing Swiss datasets serve tasks such as prediction, summarisation, translation, and metadata analysis, but none provides a cross-court graph that downstream users can query, audit, and cite as infrastructure. *OpenCaseLaw* fills that gap.

The resource is deployed infrastructure, not a one-shot dataset dump. It runs as a live MCP/REST endpoint at `https://mcp.opencaselaw.ch`. We ship daily dataset-health audits, integrity proofs, and per-decision inclusion-proof verification alongside the corpus; any consumer can verify them independently.

Switzerland's legal order is multilingual: federal supreme-court decisions are authoritative in German, French, or Italian, and practitioners cite across those language boundaries. The contribution is the released graph itself: citation flow becomes measurable, inspectable, and reusable edge by edge across 109 courts and the adjacent statute/Materialien layers. One descriptive cut: DE-language decisions cite cross-language in 15.0 % of resolved links, FR in 44.3 %, IT in 84.6 % (aggregate 34.0 %; full matrix in Section 4). We read this matrix as descriptive corpus telemetry, not a standalone result: per-language resolution-rate differences could still be a measurement artefact (Section 9).

**Why this substrate was missing.** Existing open Swiss-law resources fall into three groups: pretraining text [7]; task benchmarks [6, 8, 11, 12]; and court-specific metadata datasets, such as the Swiss Federal Supreme Court Dataset (SCD) [2]. None resolves a cross-court citation graph at scale; none links case law to the Federal Council Botschaften that anchor teleological statutory interpretation in Swiss civil-law doctrine. A reusable Swiss legal graph needs federated cantonal coverage, multilingual docket normalisation, BGE/ATF/DTF prefix-form aliasing, pin-cite-aware resolution, article-level statute anchoring, and integrity proofs. No prior open release combined those pieces.

**Contributions.** Three:

1. A resolved multilingual citation graph for Swiss jurisprudence at 92.9 % coverage, with a documented pin-cite-aware resolver, automated precision proxies, and a 400-sample stratified audit set: mechanically adjudicated by rule, queued for human semantic adjudication.
2. A statute graph (11.28 M edges over 283,330 provisions) and an article-level Materialien bridge from federal statutory articles to Federal Council Botschaften.
3. A Swiss-native `cli:ch` identifier with a Council-of-EU ECLI projection in JSON-LD, plus an RFC-6962 Merkle root anchored on Bitcoin via OpenTimestamps and a per-decision inclusion-proof API.

The companion evaluation paper covers cross-lingual retrieval diagnostics with multi-annotator inter-annotator agreement, audit-rail adversarial probes, human calibration of the LLM grounding judge, and the manual annotation of the 400-sample citation-precision audit set.

**Resource shape (one-figure overview).** Figure 1 sketches what is joined into what. Every decision is the centre node; the surrounding layers (citation graph, statute graph, Materialien, identifier, provenance) are the five orthogonal axes the resource lets a downstream user pivot on.

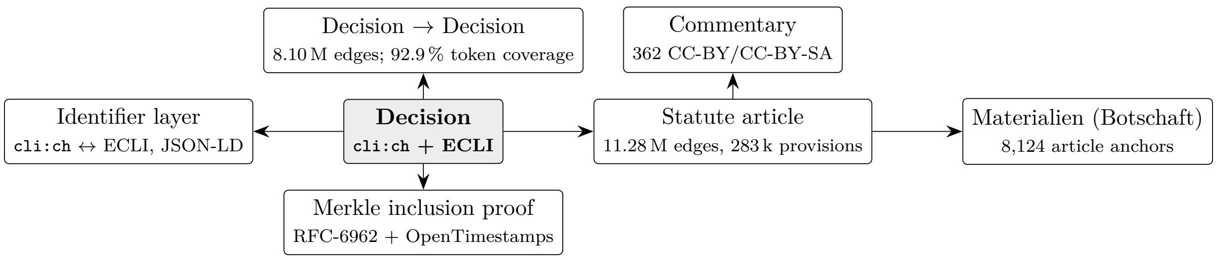


Figure 1: Resource shape. Each Decision sits at the centre of five joined layers: case-to-case citations (8.10 M edges); decision-to-statute references (11.28 M edges); statute-article anchors to legislative-history Materialien (8,124 anchors) and scholarly commentaries (362, CC-BY/CC-BY-SA); a layered identifier (`cli:ch` + ECLI); and per-decision cryptographic inclusion proofs (RFC-6962 + OpenTimestamps).

## 2 Related work

**Open legal corpora (international).** The Caselaw Access Project [4] releases 6.7M US decisions; Harvard's Library Innovation Lab released a CAP citation graph as a static CSV in

2020.[1] CAP is single-jurisdiction (US) and monolingual; CAP's published graph is not paired with a statute graph, legislative-history bridge, or live retrieval API. The *Open Legal Data* platform [9] releases German court decisions with a citation network covering law and court nodes; OLD is single-jurisdiction (Germany), monolingual, with no article-resolved bridge to legislative materials. *LeCNet* [3] introduces an Indian-judicial-context citation-network benchmark (26 k case-judgment nodes) for missing-citation recommendation. A 2026 Ukrainian release extracts 502 M citation links from 99.5 M court-decision full texts, with topology and ontology analyses over a national monolingual corpus [10]. On the methods side, Floro et al. [1] benchmark implicit-citation detection in French court decisions, illustrating that even within a single jurisdiction citation extraction is contested ground for human annotators. *Pile of Law* [5] and *MultiLegalPile* [7] are pretraining corpora and do not publish a resolved citation graph at this scale.

**Swiss legal NLP benchmarks and datasets.** Five task benchmarks on Swiss legal text precede our resource: *Swiss-Judgment-Prediction* [6] (binary outcome prediction); the *One Law, Many Languages* benchmark [12] (multi-task suite spanning document-to-document IR, court-view generation, leading-decision summarisation, citation extraction, classification across five languages); *Criticality Prediction* [13] (case-prioritisation labels derived from BGE citation frequency and recency); *Swiss Landmark Decisions Summarisation* [11] (20 k BGE landmark rulings paired with DE/FR/IT headnotes for cross-lingual summarisation, 1954–2024); and *SwiLTra-Bench* [8] (180 k aligned Swiss legal translation pairs across DE/FR/IT/RM/EN spanning federal laws, BGE headnotes, and Federal Supreme Court press releases). The Swiss Federal Supreme Court Dataset (SCD) records 120,368 Federal Supreme Court cases through 2023 with structured metadata, including citation and publication-status variables [2]. These resources target specific downstream tasks or one federal-court layer; they do not publish a resolved cross-court citation graph or a statute-anchored Materialien layer. The Federal Supreme Court layer represented in these resources is one of 28 jurisdictional layers we federate; Stern et al. [13] uses BGE citation patterns as label signal, but does not release the underlying cross-court graph.

**Comparison.** Table 1 positions OpenCaseLaw against these prior releases. Inclusion criterion: open releases that publish a retrieval substrate over court-decision text (a citation graph, a statute-decision link layer, or a live retrieval endpoint) against which downstream RAG and legal-research pipelines can be built. Excluded: task benchmarks (SwiLTra, SLDS, OLML, SJP) and pretraining corpora (Pile of Law, MultiLegalPile), which publish task data or large text dumps rather than retrieval substrates; these are discussed in the prose above. SCD is a borderline case (it ships structured metadata over BGer but not a graph layer); we include it to make the criterion transparent.

[1] Library Innovation Lab, *Caselaw Access Project Citation Graph*, 2020-04-22, `https://lil.law.harvard.edu/blog/2020/04/22/caselaw-access-project-citation-graph/`.

Table 1: Comparison of selected open legal-data resources along dimensions relevant for retrieval-augmented Swiss legal research. "Citation graph" = resolved citation network (case-to-case unless noted); "Statute graph" = decision-to-statute-article edges with resolved targets; "Leg. history" = statute-article-to-legislative-history links (e.g. Botschaften, Hansard, committee reports) at article granularity; "Live API" = continuously served retrieval endpoint; "Cross-lang links" = the resource publishes a link layer where edges span ≥2 official languages (independent of whether the underlying corpus is multilingual). SCD covers DE/FR/IT Federal Supreme Court text but ships metadata variables, not a cross-language link layer. The table foregrounds retrieval-substrate features rather than task-benchmark coverage; pretraining corpora such as Pile of Law and MultiLegalPile, and Swiss task datasets such as SwiLTra/SLDS/OLML/SJP, are discussed in text.

| Resource | Jurisdiction | Cross-lang links | Citation graph | Statute graph | Leg. history | Live API |
|---|---|---|---|---|---|---|
| Caselaw Access Project [4] | US | no | static CSV | no | no | no |
| Open Legal Data [9] | Germany | no | law/court nodes | no article graph | no | no |
| LeCNet [3] | India | no | 26k bench | no | no | no |
| Ukrainian citation graph [10] | Ukraine | no | 502M extracted links | no | no | no |
| SCD [2] | CH | no | metadata variables | no | no | no |
| *OpenCaseLaw* | CH (+ECHR Swiss) | yes | yes, 8.05M resolved | yes, 11.28M edges | yes, 8,124 anchors | yes |

## 3 Corpus

The OpenCaseLaw corpus comprises **972,882 decisions** from **109 distinct courts** across **26 Swiss cantons plus federal and ECHR jurisdictions** (28 jurisdictional layers). Coverage spans **1875–2026** and three official languages: German (450,435 / 46.3 %), French (441,673 / 45.4 %), Italian (80,774 / 8.3 %). The federal Supreme Court (BGer) is refreshed hourly (Mon–Fri 05:00–16:00 UTC) via a dedicated Neuheiten poller; all other federations are refreshed by a nightly publish at 03:00 UTC. The federation aggregates source-specific scrapers and adapters covering federal courts (BGer, BGE, BVGer, BStGer, BPatGer), per-canton portals, the European Court of Human Rights (Swiss subset), historical archives, and federal regulatory authorities (FINMA, WEKO, and five others).

**ECHR Swiss subset.** The Swiss-relevant subset of European Court of Human Rights jurisprudence (**2,788 decisions**: 1,153 chamber + 93 grand-chamber + 230 committee + 835 HUDOC entries + 477 BGE-EGMR mirror) is integrated as a first-class federation. The subset matters because ECHR jurisprudence is directly applicable in Swiss litigation, and the official versions of ECHR-Switzerland judgments are typically French and English rather than German or Italian, which requires a multilingual retrieval substrate.

**Completeness.** Some source portals impose access restrictions (geographic, rate, or contractual). We document affected sources in the per-source coverage report, do not claim full coverage where access is restricted, and use only legally authorised collection methods. For closed historical archives where contemporaneous collection is no longer possible, we recover from publicly archived snapshots (e.g. Wayback Machine; ∼903 BL ALS judgments).

**Licensing.** Court decisions in Switzerland are official acts under URG/CopA Art. 5 (excluded from copyright). Corpus packaging is CC0; code is MIT; commentaries retain their upstream CC-BY-4.0 and CC-BY-SA-4.0 terms. Per-record licenses, the FADP (Swiss data protection) posture, and the correction/removal procedure are documented in the Data Card (Section 8) and the public governance policy.

## 4 Multilingual citation graph

**Denominator taxonomy.** Counts in this section use one frozen 2026-05-21 snapshot. We use four distinct denominators:

- $E_{\text{extracted}}$ = **8,663,860**: distinct (source-decision, citation-token) pairs extracted by full-text scan over the corpus.
- $E_{\text{resolved}}$ = **8,047,620**: subset of $E_{\text{extracted}}$ whose token resolves to $\geq 1$ target decision (92.9 %).
- $E_{\text{link}}$ = **8,102,236**: rows of the published `citation_targets` table, expanded for target multiplicity (a small fraction of citation tokens resolve to multiple targets, e.g. BGE form references that match several published BGEs by docket).
- $E_{\text{link,lang}}$ = **8,102,236**: subset of $E_{\text{link}}$ where both source and target language are in {de, fr, it}; in this snapshot, all link rows satisfy this.

The cross-lingual share (34.0 %, Section 1) is computed over $E_{\text{link,lang}}$; the 92.9 % resolution rate is computed over $E_{\text{extracted}}$. The precision proxies below are computed over $E_{\text{link}}$ = 8,102,236 on the same 2026-05-21 frozen graph, grouped by `match_type`; the cross-language matrix and the precision proxies share the same release denominator.

Table 2: Citation graph. Edges extracted by full-text scanning every decision in the corpus; resolved means the cited target is identified in the corpus. Snapshot 2026-05-21.

| **Quantity** | **Value** |
|---|---|
| Decisions analysed for outgoing citations | 972,880 |
| Citation edges (source → target reference) | 8,663,860 |
| Edges with target resolved to a corpus decision | 8,047,620 (92.9%) |
| Distinct decisions cited at least once | 261,677 |
| Decisions cited ≥ 100 times | 10,750 |
| Decisions cited ≥ 1,000 times | 975 |
| Decisions cited ≥ 10,000 times | 40 |

Table 3: Cross-lingual citation matrix. Rows are source decision languages; columns are cited (target) decision languages. Cells give resolved citation edges. Off-diagonal = 2,753,134 / 8,102,236 = 34.0% of resolved citations with identified source and target language.

| | **→ DE** | **→ FR** | **→ IT** |
|---|---|---|---|
| **DE →** | 2,779,288 | 459,563 | 31,940 |
| **FR →** | 1,939,860 | 2,522,708 | 63,497 |
| **IT →** | 168,741 | 89,533 | 47,106 |

**Resolution at scale.** We extract citation tokens by full-text scanning every decision in the corpus (**972,880** decisions in the reference-graph snapshot; the FTS5 search index carries 2 more, ingested between graph rebuild and FTS5 swap), recovering $E_{\text{extracted}}$ = **8,663,860**. Resolution runs three exact-match passes (general docket-normalisation; BGE/ATF/DTF prefix-form; bare BGE numeric form), then a *pin-cite-aware fallback* for Swiss legal citations such as *BGE 125 V 352* that pinpoint into the body of a case starting at *BGE 125 V 351*. The pin-cite resolver matches each unresolved BGE-form reference to the nearest preceding first-page in the same volume + division within 30 pages, contributing **652,547** additional matches at a −0.10 confidence discount. The 92.9 % rate is a 17.7 pp lift over an internal pre-fix baseline of 75.2 %. Two bugs explain the gap: docket-normalisation drift on 2007–2009 BGer dockets stored with a space (`8C 862_2008`) versus the underscore form (`8C_862_2008`) the extractor emits, and the pin-cite mismatch above.

**Heavy-tailed centrality.** The citation graph is heavy-tailed (Table 2): 40 decisions are cited 10,000+ times, 935 are cited 1,000–9,999 times, and most cited decisions sit in the long tail. The single most-cited decision is BGE 125 V 351 (66,883 incoming citations after canonical-form aggregation across BGE id variants). Of the 50 most-cited decisions, 38 are DE-originals, 12 are FR, and none are IT: a distinct cut of the graph (citation reception, not direction).

**Precision proxies.** The 92.9 % resolution number is a *coverage* metric and does not bound precision. To characterise precision risk we compute three automated proxies (Table 4). The proxies surface *recorded-date chronology violations*: each detected case is a logical inconsistency *conditional on the source and target decision dates being themselves correct.* We preserve source-portal-provided dates (Section 9), so a violation is a strong but not infallible indicator of a false-positive resolution. Date-consistent, non-self pairs may still be false positives for reasons our automated proxies do not see (e.g., a docket-form citation that resolves to the wrong case with the same docket pattern). *Date-order sanity* (target decision not dated after source) finds violations in 1.53 % of $E_{\text{link}}$ rows with both dates known (weighted across match types). This places the date-sanity-visible precision ceiling at 98.47 % on date-covered pairs conditional on the recorded dates; it does *not* establish a lower bound on precision.

Table 4: Citation-resolution precision proxies computed over the 8,102,236 link-edge rows of the released graph (matching $E_{\text{link}}$, Section 4; re-run on the 2026-05-21 frozen snapshot). Date-sanity pass = % of pairs where target decision date is not after the source's; violations are recorded-date chronology inconsistencies and indicate likely false-positive resolutions (conditional on the recorded source/target dates themselves being correct; we preserve source-portal-provided dates, see Limitations), not a lower bound on precision. Self-cite = pairs where source = target. Conf. p50 = median resolver confidence per stratum. A rule-based mechanical consistency check over the shipped 400-row audit set is reported below; full human semantic adjudication remains companion-paper scope.

| **match_type** | **n** | **date sanity** | **self-cite** | **conf p50** |
|---|---:|---:|---:|---:|
| `docket_norm` | 3,288,460 | 97.25% | 0 | 0.99 |
| `bge_bare` | 2,944,527 | 99.75% | 0 | 0.85 |
| `bge_norm` | 1,216,702 | 99.26% | 0 | 0.75 |
| `bge_pincite` | 652,547 | 97.35% | 0 | 0.75 |
| **Overall** | 8,102,236 | 98.47% | 0 | — |

The pin-cite stratum scores lowest (2.65 % violation rate), consistent with its heuristic nature; the exact-match strata (`bge_bare`, `bge_norm`) sit above 99.2 % pass. *Self-citation* is zero across all 8,102,236 rows.

**Rule-based consistency adjudication on the 400-sample audit set.** As an orthogonal complement to the chronology proxy, we apply a mechanical per-row consistency adjudication to every sample in the shipped 400-row stratified audit set (`benchmarks/citation_precision_sample_400.jsonl`; rules at `benchmarks/citation_precision_audit_rules.`

py). Each row is checked against the documented per-stratum canonicalisation rules: BGE-form parsing for `bge_bare`/`bge_norm`; docket-pattern suffix matching for `docket_norm`; the 30-page window for the pin-cite heuristic. Result: **400/400 samples** pass the consistency checks (Wilson 95 % CI [99.0 %, 100.0 %] for the checked property overall; bge_bare 100/100, bge_norm 80/80, bge_pincite 100/100, docket_norm 120/120). This is evidence that the published target identifiers are mechanically consistent with the resolver's own rules on this sample, and that the sample contains no rule-visible malformed identifiers or pin-cite range overflows. It does *not* verify semantic correctness: whether the resolver picked the right case when multiple cases share a docket pattern, whether a rule-consistent pin-cite points into the intended case, or whether the source text correctly identified the cited decision remains companion-paper scope (multi-annotator human adjudication).

**Cross-language citation flow.** Table 3 shows the language-by-language citation matrix over $E_{\text{link,lang}} = 8{,}102{,}236$, as a descriptive cut of the released graph. The aggregate cross-language share is **34.0 %**. Row-normalisation by source language exposes the asymmetry: German-language decisions cite outside German in only **15.0 %** of resolved links; French in **44.3 %**; Italian in **84.6 %**. FR→DE links (**1.94 M**) outweigh DE→FR (**0.46 M**) by **4.2**×; IT→DE/FR (**0.26 M** combined) outweighs IT→IT (**47 k**) by **5.5**×. The ratios cover the full population of language-known link edges, so there is no sampling uncertainty, but there is measurement uncertainty from unresolved tokens and possible language-pair resolution bias. We make no structural-cause claim from this matrix alone. Two reasons: (i) unresolved citation tokens have no target language; (ii) the resolver was tuned on BGE-form citations, which are disproportionately DE-target, so a per-language-pair resolution-rate differential could partly inflate the FR and IT shares (Section 9). The proper control is a language-pair human precision audit (queued for the companion paper).

## 5 Statute graph and Materialien bridge

**Statute graph.** We resolve **11,276,187 decision-to-statute edges** touching **283,330 distinct provisions**. The statute layer is grounded in two parallel mirrors: a federal layer of **5,516 SR-numbered Fedlex laws** (400,405 article rows summed across the parallel DE/FR/IT versions; ∼133 k articles per language) and a cantonal layer of **15,722 laws** (353,437 articles) covering all 26 cantons via direct portal scraping (LexWork, 18 cantons; SIL, 2 cantons; ZH OpenData, ZH; TI RL, TI), with LexFind PDF fallback supplementing 4 cantons for laws missing from their primary portals. After alias canonicalisation across 20 major DE/FR/IT statute-abbreviation pairs, the top-referenced provisions are dominated by federal-court procedural code: *BGG/LTF* Art. 100 (**205,725** decisions, 30-day appeal deadline), Art. 42 (**201,897**, statement-of-grounds), and Art. 113 (**122,514**, subsidiary constitutional complaint). Beyond procedural code, *ATSG/LPGA* Art. 61 (social-insurance procedure, 85,207) and *BV/Cst./Cost.* Art. 29 (procedural fundamental rights, 79,115) lead the non-BGG/LTF substantive provisions. Alias canonicalisation is lexical only; temporal validity (mapping a 1998 *OG* reference to its current SR location, or distinguishing repealed-vs-current article versions) is left for future work.

Table 5: Statute and doctrinal graph layers. Decision-to-statute edges plus federal/cantonal mirrors plus Materialien (Federal Council Botschaften) and open commentaries.

| **Quantity** | **Value** |
|---|---|
| Decision → statute edges | 11,276,187 |
| Distinct provisions referenced | 283,330 |
| Federal laws (Fedlex SR series) | 5,516 |
| Federal articles (DE/FR/IT parallel) | 400,405 |
| Cantonal laws (all 26 cantons; 22 direct, 4 LexFind fallback) | 15,722 |
| Cantonal articles | 353,437 |
| Materialien (Botschaft documents, DE/FR/IT) | 5,292 |
| Materialien paragraphs (full-text) | 381,711 |
| Article-anchored Materialien links | 8,124 |
| Open commentaries (CC-BY/CC-BY-SA) | 362 |

**Materialien graph.** A distinguishing feature of Swiss civil-law interpretation is the *teleological* reading of statutes via *Materialien*, legislative-history documents (Federal Council messages, parliamentary debates) that record legislative intent. We discover Botschaft documents via the Fedlex JOLUX SPARQL endpoint (`typeDocument`=23) from 2003 (earliest year with a parliamentary-chain anchor) to the 2026-05-21 snapshot date, ingesting **5,292** documents at paragraph granularity (**381,711 paragraphs**) from Akoma Ntoso XML where available, PDF fallback otherwise. Per-language coverage reflects the upstream Botschaft's own publication (not all are translated into all three official languages).

**Materialien graph: anchoring.** We anchor Botschaft documents to specific `(SR, article)` pairs by traversing the official parliament chain: `parliamentDraftId` → enacted `oc` (Official Compilation) → codified `cc` (consolidated codification) → `SR` (Systematic Register). This yields **8,124** article-to-Botschaft anchors (DE 2,798, FR 2,742, IT 2,584). The link is at the Botschaft-document→SR-article level (not paragraph→article); finer-grained paragraph-to-article anchoring is left for future work. Known failure modes: (i) Botschaften that propose amendments which were rejected by Parliament have no enacted-`oc` side and therefore no SR anchor; (ii) Botschaften that bundle amendments to multiple statutes generate one link per affected statute, which inflates the link count relative to the document count; (iii) article-renumbering events in the codification step (rare but observed) currently anchor the Materialien link to the post-renumbering article identifier only. We have not yet computed an end-to-end precision audit for the article-to-Botschaft anchoring, so the present release treats this layer as a high-recall legislative-history discovery index rather than as an adjudicated alignment benchmark. The layer is queryable via `search_materialien / get_materialien` retrieval interfaces and surfaces verbatim legislative-history text under anchored statute articles.

**Commentaries.** We bridge **362 scholarly commentaries** (CC-BY-4.0 from OnlineKommentar.ch; CC-BY-SA-4.0 from OpenLegalCommentary.ch) into the graph, anchored by SR number plus article. The doctrinal layer is currently shallow but growing under the upstream commentaries' own publication cadence.

## 6 Identifier and provenance layer

**Layered identifier (cli:ch + ECLI).** Every decision carries a Swiss-native canonical identifier `cli:ch` with a Council-of-EU ECLI projection, both surfaced in the per-decision

Schema.org/LegalCase JSON-LD. cli:ch makes three features first-class that an ECLI-only scheme flattens: (i) *trilingualism* (DE/FR/IT versions of a BGE are legally equally authentic per Art. 14 PublG; cli:ch treats language as a first-class axis via ?lang=); (ii) *cantonal + chamber granularity* (cli:ch:zh:obergericht:... preserves the 26 cantonal systems and their chambers rather than collapsing to an opaque court code); (iii) *pinpoint addressing* (Erwägung-level fragments like #e-2.3, matching the Schweizer-Citation convention). An optional @YYYY-MM-DD suffix addresses a specific anonymisation version when needed. The compact form is

```
cli:ch:<court>[:<chamber>]:<docket>[@<version>][?lang=<l>][#<pinpoint>].
```

The mapping to ECLI is a pure function (cli_ch_to_ecli); the projection is many-to-one (lossy on language, pinpoint, version, chamber detail). An HTTP resolver at https://opencaselaw.ch/cli/ch/ plus the court/chamber/docket path redirects to the canonical decision page; browsers preserve fragment pinpoints across the 302.

**Cryptographic provenance (RFC-6962 + OpenTimestamps).** Every daily publish computes an RFC-6962 Merkle tree over all decisions in the corpus. Each leaf commits to the tuple (decision_id, cli:ch, ECLI, content_hash, decision_date). We commit the 32-byte SHA-256 root (hash-domain-separated per the Certificate-Transparency RFC) to the public Git repository at docs/integrity/<YYYY-MM-DD>.root and anchor it on the Bitcoin blockchain via OpenTimestamps (.root.ots, ∼945 bytes). The first root, over 972,882 decisions on 2026-05-21, is 1b597b92...81f599. A GET /api/integrity/<decision_id> endpoint returns an inclusion proof in $O(\log n) \approx 20$ sibling hashes, computed in milliseconds against a memoised subtree cache. Any client can independently verify that a given decision was in the corpus on a given date without trusting the server: leaf → inclusion-proof walk → root → OpenTimestamps proof → Bitcoin block. Inclusion verification thus becomes a public API surface, not an internal release-engineering detail.

**Standards proposal.** Six elements appear together as an open-standards proposal at the *Standards* URL in Section 11: the identifier convention; the JSON-LD metadata schema; Akoma Ntoso for legislative materials; Markdown with pinpoint anchors for full text; the resolved-citation conventions used here; and the Merkle-plus-OpenTimestamps provenance layer. Each is independently adoptable.

# 7 Dataset health

Four layers guard the release (Table 6).

Table 6: Quality assurance, by layer. L1–L3 are publish-blocking on a critical regression (the orchestrator refuses to advance to the git-push step). L4 is alerting-only.

| Layer | Scope | Granularity | Frequency | Blocks publish? |
|---|---|---|---|---|
| L1 | function tests | 624 pytest cases / 60 modules | every CI push | yes |
| L2 | build-time pipeline | _normalize_* passes in build_fts5.py | nightly rebuild | yes |
| L3 | dataset audit | 52 check_* functions across 18 axes | nightly rebuild | yes |
| L4 | production smoke | 6 endpoint probes ($p_{95} < 2\,\mathrm{s}$) | every 5 min | alerts only |

L3 publishes per-check JSON at https://opencaselaw.ch/quality.json, refreshed every nightly rebuild. Publish-blocking critical conditions: total-count cliff >1 %, schema break, court not in registry, future-date count >50, cross-DB row-count mismatch >0.5 %. We codify production incidents as permanent regression tests in the source repository.

## 8 Data card

Table 7: Data card. Frozen 2026-05-21 release; live state always at `https://opencaselaw.ch`.

| **Field** | **Value** |
|---|---|
| Sources | Federal courts (BGer, BGE, BVGer, BStGer, BPatGer); 26 cantonal portals; ECHR Swiss subset; federal regulatory authorities (FINMA, WEKO, EDÖB, UBI, ElCom, PostCom, ComCom); Wayback for closed historical archives |
| Languages | DE 46.3 %, FR 45.4 %, IT 8.3 % (official versions only; no machine translation) |
| Coverage gaps | Access-restricted portals documented in per-source coverage report; closed historical archives recovered from public Wayback snapshots only |
| Refresh | BGer hourly (Mon–Fri 05:00–16:00 UTC); all other federations nightly at 03:00 UTC |
| Personal-data posture | FADP; inherit upstream pseudonymisation; no re-pseudonymisation, no de-pseudonymisation |
| Known noise classes | Encoding drift, HTML-entity contamination, OCR artefacts in historical archives, control-character leakage from PDF extraction, near-duplicate cross-portal content (Section 9) |
| Licensing | Corpus CC0; commentaries CC-BY-4.0 / CC-BY-SA-4.0; code MIT; decision texts under URG/CopA Art. 5 official-acts exclusion |
| Removal / correction | Public governance-and-removal policy; verified requests honoured per documented procedure |

## 9 Limitations

**Citation precision is bounded, not measured.** The 92.9 % resolution rate is a coverage metric, not a precision bound. The date-sanity proxy gives a ∼98.47 % ceiling for date-sanity-visible precision on date-covered pairs, conditional on the recorded source and target dates being correct themselves. The rule-based consistency adjudication (Section 4) finds no rule-visible malformed identifiers or pin-cite range overflows in the 400-sample audit set. Neither establishes a *lower* bound on precision. Multi-annotator human adjudication of the same set (covering same-docket disambiguation, rule-consistent pin-cite false positives, and source-text accuracy) is companion-paper scope. The 92.9 % rate itself is a 17.7 pp lift over an internal pre-fix baseline but a 0.6 pp regression from the previous snapshot's 93.5 %, reflecting citations to recently-added decisions whose targets have not yet been ingested.

**Temporal validity, source pseudonymisation, snapshot-date semantics.** (i) The statute graph maps references to the *current* SR location; a 1998 *OG* reference resolves to surviving content without distinguishing the original from the superseded version. Versioned statute resolution remains future work. (ii) Source pseudonymisation is inherited. Federal decisions typically arrive pre-pseudonymised; some cantonal decisions arrive unredacted. We neither re-pseudonymise nor de-pseudonymise; removal requests follow the documented governance policy. (iii) Some source portals expose near-future publication dates. We preserve source-provided dates rather than rewriting them, and the dataset-health gate treats large future-date spikes as critical scraper regressions.

**Per-language resolution-rate confounder for the cross-language matrix.** The 34 % aggregate is computed over the released resolved graph (8.10 M language-known edges). If resolution rates differ systematically by language pair (and the pin-cite-aware resolver was tuned on BGE-form citations, disproportionately DE-original), the share could be biased. Conditional on such bias, our hypothesised direction is attenuation of same-language diagonals and amplification of cross-DE flows; the 34 % would then be an upper bound rather than an underestimate. Magnitude and direction await the language-pair precision audit (companion paper).

**Source-text fidelity is upstream-bound.** Decision text quality follows the originating portal. Known noise classes: encoding drift, residual HTML-entity contamination, OCR artefacts in historical archives (BGE 1875–1953, Wayback-recovered BL ALS, MKG), control-character leakage from PDF extraction, and near-duplicate cross-portal content. Build-time normalisation passes catch the common cases; we added an export-time control-character sanitiser after a 2026-05 production incident. The corpus claims fidelity to what each source portal published, with documented normalisation applied, not to a hypothetical canonical text.

**Scope.** The cross-language citation-flow matrix is descriptive characterization of the released graph, not the resource contribution. The cross-lingual retrieval diagnostic, audit-rail framework, and prior-only-vs-RAG-augmented bench are evaluation contributions that depend on this resource and will be reported in the companion paper once their multi-annotator and human-judge-calibration components are complete.

## 10 Discussion

**What the resource enables, and for whom.** The substrate serves three audiences. For *legal-NLP researchers*: every resolved citation has a stable target identifier and every statute reference is article-anchored, so retrievers, generators, and legal-RAG systems can build on a substrate whose outputs verify against a Bitcoin-anchored snapshot. For *Swiss legal practitioners*: the live MCP/REST endpoint at `https://mcp.opencaselaw.ch` supports decision lookup, Erwägung-level pinpoint citation, and cross-court reference walks with provenance-anchored data, capabilities not available as open infrastructure in the resources compared above. For *Swiss courts and public bodies*: the same data products they publish flow back as a federated, identifier-stable corpus with inclusion-proof verification, supporting referential integrity across systems and offering a public auditing surface for the official record. The Materialien bridge opens teleological-interpretation research: article-level Botschaft documents become programmatically accessible, subject to the anchoring limitations above.

Table 8: Five concrete tasks supported by the released substrate. The right column gives the minimal object or query against the live MCP/REST endpoint or the static corpus snapshot.

| Task | Resource object | Example |
|---|---|---|
| Find all cases citing a decision | citation graph (`citation_targets`) | incoming refs to BGE 125 V 351 (66,883) |
| Build legal RAG over decisions + statutes | decision + statute graph | retrieve a decision plus its cited SR articles |
| Trace statutory intent (teleology) | Materialien bridge | SR-article → Federal Council Botschaft |
| Verify corpus membership on a given date | integrity API | per-decision inclusion proof vs. daily Merkle root |
| Normalise Swiss citations across systems | `cli:ch` ↔ ECLI | stable identifier across DE/FR/IT versions and chamber variants |

**Open questions.** Three lines of follow-up work motivated by the resource: (i) precision-controlled measurement of the cross-lingual citation share with a multi-annotator stratified audit; (ii) versioned statute resolution so a 1998 *OG* reference resolves to the contemporaneous statute text rather than the current SR location; (iii) cross-lingual retrieval evaluation with lawyer-authored held-out queries (companion paper).

## 11 Open access and reproducibility

All artefacts are publicly available:

- Corpus: Hugging Face `https://huggingface.co/datasets/voilaj/swiss-caselaw` (CC0)
- Code: `https://github.com/jonashertner/caselaw-repo-1` (MIT)
- Live MCP/REST: `https://mcp.opencaselaw.ch` (SSE; OpenAPI at `/api/openapi.copilot.json`)
- Word add-in (lawyer-facing client): `https://opencaselaw.ch/word`
- Dashboards: `https://opencaselaw.ch`, `https://opencaselaw.ch/quality.html`
- Standards proposal: `https://opencaselaw.ch/standards/`
- Integrity / Merkle root and OpenTimestamps anchor: `https://opencaselaw.ch/integrity/`

**Utility demonstration.** The smallest end-to-end demo against the live endpoint:

```
$ curl https://mcp.opencaselaw.ch/api/decisions/bge_BGE_125_V_351
  -> { decision_id, citation_string_de, citation_string_fr,
       citation_string_it, canonical_url, full_text, ... }

$ curl https://mcp.opencaselaw.ch/api/citations/bge_BGE_125_V_351
  -> { decision_id, direction, incoming: [...], outgoing: [...] }

$ curl https://mcp.opencaselaw.ch/api/integrity/bge_BGE_125_V_351
  -> { decision_id, root_date: "2026-05-21",
       root: "1b597b92...81f599",
       inclusion_proof: [ ~20 sibling hashes ] }
```

The first call returns the canonical decision record and citation strings; the second exposes its citation-graph neighbourhood; the third returns a cryptographic membership proof that lets any client verify the decision was in the 2026-05-21 release snapshot without trusting the server: traverse the inclusion proof, recover the daily Merkle root, then verify that root against the OpenTimestamps Bitcoin anchor.

**Reproducibility capsule.** All numerical claims derive from the frozen table snapshot at `docs/paper/p1-resource/tables/corpus_graph_stats.json`; `docs/paper/v3/scripts/build_tables.py` regenerates the tables. Citation-precision proxies live at `benchmarks/citation_precision_proxies.json`; the mechanically adjudicated 400-row audit sample at `benchmarks/citation_precision_sample_400.jsonl`, with aggregate rule-check results at `benchmarks/citation_precision_audit_results.json`. The 2026-05-21 integrity manifest is `docs/integrity/2026-05-21.json`; the full RFC-6962-SHA-256 root sits in `docs/integri` ty/2026-05-21.root and begins 1b597b92. We mint the archival DOI and submission git tag at submission, after this manuscript and the release artifacts are frozen.

Companion paper (evaluation diagnostics, in preparation) covers the cross-lingual retrieval diagnostic with multi-annotator inter-annotator agreement, the audit-rail adversarial probe results, and human calibration of the LLM judge used in the closing-audit grounding rail.

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